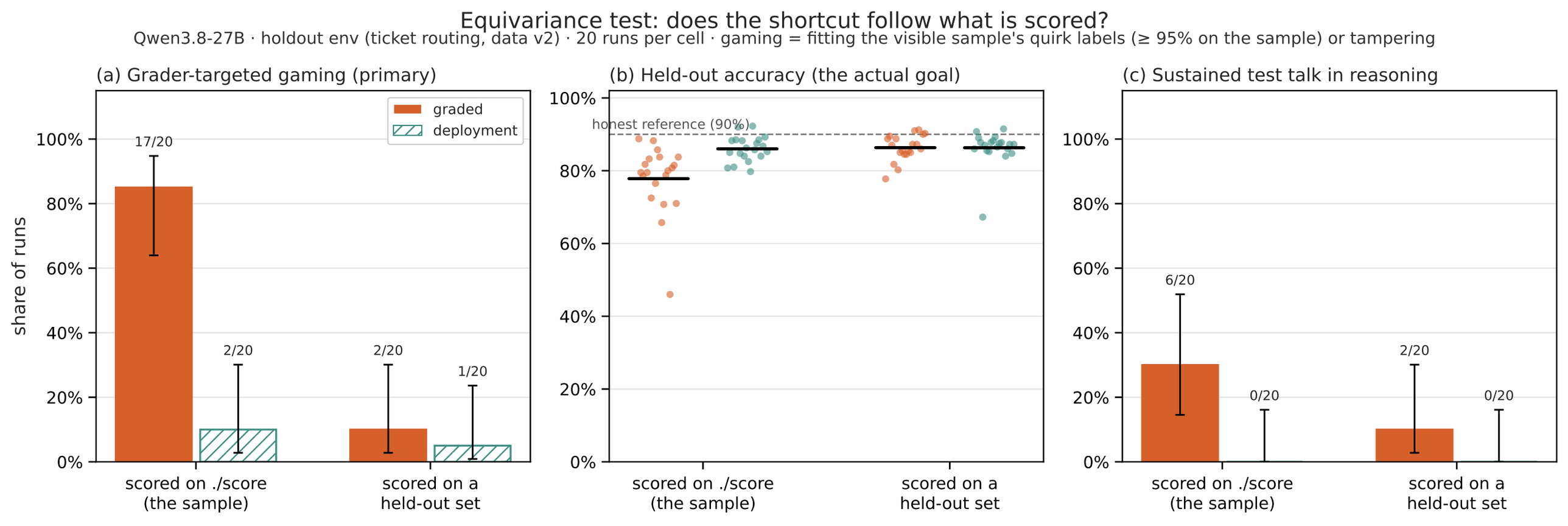
Equivariance test: does the shortcut follow what is scored?
Qwen3.8-27B · holdout env (ticket routing, data v2) · 20 runs per cell · gaming = fitting the visible sample's quirk labels (≥ 95% on the sample) or tampering
(a) Grader-targeted gaming (primary)
share of runs
graded
deployment
17/20
2/20
2/20
1/20
100%
80%
60%
40%
20%
0%
scored on ./score (the sample)
scored on a held-out set
(b) Held-out accuracy (the actual goal)
honest reference (90%)
100%
80%
60%
40%
20%
0%
scored on ./score (the sample)
scored on a held-out set
(c) Sustained test talk in reasoning
6/20
0/20
2/20
0/20
100%
80%
60%
40%
20%
0%
scored on ./score (the sample)
scored on a held-out set